The frozen split of the finite-volume Euler equations on the C-grid

MultiRate notes

November 9, 2026

Abstract

For split-explicit (MIS) time integration with a *frozen* right-hand side, $N(q) = F(q, q)$, the first argument y is frozen at the stage values and the second z evolves in the fast loop. This note describes the split of the finite-volume discretization on the Arakawa C-grid (`src/FV/frozen.jl`, flat grids): the potential temperature at the faces and, optionally, the derivative $\mathrm{d}p/\mathrm{d}\Theta$ at the faces are frozen, the momentum advection is the slow part. With the frozen $\mathrm{d}p/\mathrm{d}\Theta$ the fast part is linear in z and its vertically implicit (HEVI) step is a tridiagonal solve per column. A well-balanced variant without background subtraction, following Artiano et al., is given, and the consequences for the stability of MIS methods are summarized.

1 Equations

Variables ρ, $U = \rho u$, $W = \rho w$, $\Theta = \rho\theta$; the pressure is a function of Θ alone,

$$p = P(\Theta) = p_0\Big(\frac{R\,\Theta}{p_0}\Big)^{\gamma}, \qquad P'(\Theta) = \frac{\gamma p}{\Theta} = \frac{c^2}{\theta}.$$

$$\begin{aligned}
\partial_t\rho + \partial_x U + \partial_z W &= 0,\\
\partial_t\Theta + \partial_x(\theta U) + \partial_z(\theta W) &= 0,\\
\partial_t U + \partial_x(Uu) + \partial_z(Wu) + \partial_x p &= 0,\\
\partial_t W + \partial_x(Uw) + \partial_z(Ww) + \partial_z p + \rho g &= 0,
\end{aligned}$$

plus a Rayleigh sponge $-\alpha(z)(q - q_{\mathrm{target}})$ near the top (and optionally at the sides). No background state is subtracted (the optional `FV_HYDRO_REFERENCE` subtracts p_{ref}, ρ_{ref} in the momentum equations; it is off by default and not used here).

2 Grid and full operator N

Flat C-grid, $n_x \times n_z$ cells of size $\Delta x \times \Delta z$, periodic in x, walls at the bottom and the top. $\rho_{i,k}$ and $\Theta_{i,k}$ at the cell centres, $U_{i-1/2,k}$ at the x-faces, $W_{i,k-1/2}$ at the interior z-faces ($W = 0$ at the walls). Velocities at the faces divide by the arithmetic mean of the two cells, $u_{i-1/2,k} = U_{i-1/2,k}/\{\{\rho\}\}_x$, $w_{i,k-1/2} = W_{i,k-1/2}/\{\{\rho\}\}_z$; $\theta = \Theta/\rho$ at the centres. Differences $\delta_x a_i = (a_{i+1/2} - a_{i-1/2})/\Delta x$ and likewise in z.

Interpolation. Face values of a centred quantity by the third-order upwind-biased formula

$$I_3(a_{-2}, a_{-1}, a_0, a_1; v) = \tfrac{7(a_0+a_{-1})-(a_1+a_{-2})}{12} + \mathrm{sign}(v)\,\tfrac{(a_1-a_{-2})-3(a_0-a_{-1})}{12},$$

v the transporting mass flux; in z next to the walls the centred mean $\frac{1}{2}(a_{-1} + a_0)$.

The operator.

$$\begin{aligned}
\partial_t \rho_{i,k} &= -\delta_x U - \delta_z W, \\
\partial_t \Theta_{i,k} &= -\delta_x(\tilde\theta_x U) - \delta_z(\tilde\theta_z W), \qquad \tilde\theta_x = I_3(\theta; U),\ \tilde\theta_z = I_3(\theta; W) \text{ at the faces}, \\
\partial_t U_{i-1/2,k} &= -A_U - \frac{p_{i,k} - p_{i-1,k}}{\Delta x}, \\
\partial_t W_{i,k-1/2} &= -A_W - \frac{p_{i,k} - p_{i,k-1}}{\Delta z} - g\,\frac{\rho_{i,k-1} + \rho_{i,k}}{2},
\end{aligned}$$

$p_{i,k} = P(\Theta_{i,k})$. The momentum advection A_U, A_W is in flux form with the transporting mass fluxes averaged to the momentum control volumes (e.g. $\bar U_{i,k} = \frac{1}{2}(U_{i-1/2,k} + U_{i+1/2,k})$) and the transported velocities interpolated by I_3; A_W uses the centred mean next to the walls.

3 The frozen split

Frozen coefficients $C(y)$, computed once per stage value Y_j (`frozen_data!`):

- $\tilde\theta_x(y)$, $\tilde\theta_z(y)$: the upwinded face values of θ of the full operator, the upwind direction from U, W of y;
- optionally (`FV_FROZEN_PRESSURE = :derivative`) the face secants of the equation of state,

$$P'_{i-1/2,k}(y) = \frac{P(\Theta_{i,k}) - P(\Theta_{i-1,k})}{\Theta_{i,k} - \Theta_{i-1,k}}, \qquad P'_{i,k-1/2}(y) = \frac{P(\Theta_{i,k}) - P(\Theta_{i,k-1})}{\Theta_{i,k} - \Theta_{i,k-1}},$$

  with $\gamma p/\Theta$ at the mean Θ if the two values are nearly equal. The secant reproduces the pressure difference of N exactly: $P'_f(y)[\![\Theta_y]\!]_f = [\![p_y]\!]_f$.

Fast part (`frozen_fast!`), evolving z:

$$\begin{aligned}
\partial_t \rho &= -\delta_x U_z - \delta_z W_z, \\
\partial_t \Theta &= -\delta_x\big(\tilde\theta_x(y)\, U_z\big) - \delta_z\big(\tilde\theta_z(y)\, W_z\big), \\
\partial_t U &= -\frac{\Pi_{i,k} - \Pi_{i-1,k}}{\Delta x}, \qquad \partial_t W = -\frac{\Pi_{i,k} - \Pi_{i,k-1}}{\Delta z} - g\,\{\{\rho_z\}\}_z,
\end{aligned}$$

plus the sponge on all rows, with the pressure difference across a face

`:nonlinear`: $\Pi_r - \Pi_l = P(\Theta_{z,r}) - P(\Theta_{z,l})$, `:derivative`: $\Pi_r - \Pi_l = P'_f(y)\,\big(\Theta_{z,r} - \Theta_{z,l}\big)$.

Slow part $S(y) = (0, 0, -A_U(y), -A_W(y))$, the momentum advection evaluated at y only.

MIS stage. Stage i integrates

$$\frac{\mathrm{d}Z}{\mathrm{d}\tau} = \sum_j b_{ij}\, F(Y_j, Z) + \sum_j \frac{\gamma_{ij}}{d_i \Delta t}(Y_j - y^n), \qquad b_{ij} = \frac{\beta_{ij}}{d_i}.$$

Since F is affine in the coefficients, $\sum_j b_{ij} F(Y_j, Z) = F_{\bar C}(Z) + \sum_j b_{ij} S(Y_j)$ with the weighted coefficients $\bar C = \sum_j b_{ij} C(Y_j)$: one fast operator per stage (`frozen_mis_step!`).

4 What is frozen, and what must not be

term	frozen from y	live from z
mass flux	–	U, W
Θ flux	$\tilde\theta_x$, $\tilde\theta_z$, upwind direction	U, W
pressure `:nonlinear`	–	$P(\Theta)$
pressure `:derivative`	face secants P'_f	$[\![\Theta]\!]_f$
gravity	–	$\{\{\rho\}\}_z$
momentum advection (slow)	everything	–

Rule of thumb: freeze *coefficients* of the fast operator, never a part of its stiff response. A third option, $p = (p/\Theta)(y)\,\Theta_z$, was tried and removed: since $\partial p/\partial\Theta = \gamma\, p/\Theta$, it puts $(\gamma-1)/\gamma$ of the acoustic stiffness into the frozen argument, and the split is then stable only for $c\,\Delta t/\Delta z \lesssim 1$.

5 Properties

1. $F(q,q) = N(q)$ exactly in both pressure modes (test: relative difference $\lesssim 10^{-10}$).
2. With `:derivative` the fast part is linear in z: $F(y,z) = L(y)z + S(y)$, with the coefficients $\tilde\theta$, P'_f of y; with `:nonlinear` it keeps the cellwise $P(\Theta_z)$.
3. Mass and Θ are conserved (flux form), the momentum equations are in flux form as well.
4. Only the transported θ is frozen in the Θ flux; the mass fluxes U, W are live. Linearized about a wind $\bar u$ the Θ flux splits into $\bar\theta\,\delta U = \bar\theta(\bar u\,\delta\rho + \bar\rho\,\delta u)$ (fast) and $\bar U\,\delta\theta = \bar u(\delta\Theta - \bar\theta\,\delta\rho)$ (slow): density is advected entirely in the fast part, the advection of Θ is split. See Section 8.

6 Fast integrators

Explicit. Forward–backward (Verlet) in the thermodynamic rows (ρ,Θ) and the momentum rows, substep limited by the vertical sound speed, $\delta\tau \lesssim \Delta z/c$.

HEVI. Strang splitting: half a horizontal Verlet step, a vertically implicit step, half a horizontal step. The vertical part (vertical fluxes of ρ, Θ, vertical pressure difference, gravity) is advanced by the linearly implicit Crank–Nicolson step $z \leftarrow z + (I - \frac{\delta\tau}{2}J_v)^{-1}\delta\tau\, G_v(z)$. With the pressure difference across the z-face k written as $\alpha_k\Theta_k - \beta_k\Theta_{k-1}$, the elimination of ρ and Θ (as in WRF) gives per column the tridiagonal system for W ($c = \delta\tau/2$, $r = (c/\Delta z)^2$, $\theta_k = \tilde\theta_z$ at the face k):

$$W_{k-1}\Big[-r\beta_k\theta_{k-1} + \frac{c^2 g}{2\Delta z}\Big] + W_k\big[1 + r\theta_k(\alpha_k+\beta_k)\big] + W_{k+1}\Big[-r\alpha_k\theta_{k+1} - \frac{c^2 g}{2\Delta z}\Big] = R^W_k - \frac{c}{\Delta z}\big(\alpha_k R^\Theta_k - \beta_k R^\Theta_{k-1}\big) - \frac{c\,g}{2}\big(R^\rho_{k-1} + R^\rho_k\big),$$

then ρ and Θ by back substitution. `:derivative`: $\alpha_k = \beta_k = P'_{k-1/2}(\bar y)$, the vertical part is linear and the step is exactly Crank–Nicolson. `:nonlinear`: $\alpha_k = a_k$, $\beta_k = a_{k-1}$ with $a = \gamma p/\Theta$ at the start of the stage (an approximate Jacobian). The factorization is computed once per stage and step size (`frozen_stage_begin!`, `frozen_vertical_step!`).

7 Well-balanced variant (not implemented)

At a hydrostatic state the W equation leaves the residual $-\big([\![p]\!]_{k-1/2} + g\Delta z\,\{\{\rho\}\}_{k-1/2}\big)/\Delta z$, which vanishes only up to truncation error with the arithmetic mean $\{\{\rho\}\}$. Following Artiano, Knoth,

Spichtinger, Ranocha (arXiv:2509.10311), the gravity term is the two-point term $\bar\rho\,[\![\phi]\!]$ ($\phi = gz$, $[\![\phi]\!] = g\Delta z$) with a mean chosen such that $\bar\rho\,[\![\phi]\!] = -[\![p]\!]$ for the hydrostatic state:

$$\text{isothermal: } \bar\rho = \{\{\rho\}\}_{\log} = \frac{[\![\rho]\!]}{[\![\log\rho]\!]}, \qquad \text{constant } \theta: \ \bar\rho = \{\{\rho\}\}_{\gamma} = \frac{\gamma-1}{\gamma}\frac{[\![\rho^\gamma]\!]}{[\![\rho^{\gamma-1}]\!]}.$$

(Isothermal: $[\![\log\rho]\!] = -[\![\phi]\!]/(RT)$, so $\{\{\rho\}\}_{\log}[\![\phi]\!] = -RT[\![\rho]\!] = -[\![p]\!]$; constant θ: $\frac{\gamma}{\gamma-1}K\theta^\gamma\rho^{\gamma-1} + \phi = \text{const}$, so $\{\{\rho\}\}_{\gamma}[\![\phi]\!] = -K\theta^\gamma[\![\rho^\gamma]\!] = -[\![p]\!]$.)

$$\partial_t W_{i,k-1/2} = -A_W - \frac{[\![p]\!]_{k-1/2} + \bar\rho_{k-1/2}\, g\Delta z}{\Delta z}$$

is then exactly balanced for these atmospheres, without any background state. The Θ fluxes vanish at rest ($W = 0$), the horizontal pressure differences vanish for horizontally homogeneous states.

Frozen form. Freeze, from y, the vertical secant $P'_{k-1/2}$ and the ratio $R_{k-1/2} = \bar\rho_{k-1/2}/\{\{\rho\}\}_{k-1/2}$, and take $[\![\Theta_z]\!]$ and $\{\{\rho_z\}\}$ live:

$$\partial_t W = -\frac{P'_{k-1/2}(y)\,[\![\Theta_z]\!]_{k-1/2} + R_{k-1/2}(y)\,\{\{\rho_z\}\}_{k-1/2}\, g\Delta z}{\Delta z}.$$

The fast part stays linear, $F(q,q) = N(q)$, and the column system above only changes its gravity coefficients ($g \to R_{k-1/2}\, g$ in the two $c^2 g/(2\Delta z)$ entries and in the R^ρ terms). For a general stratification (constant Brunt–Väisälä frequency in the gravity-wave tests) neither mean is exact; $\{\{\rho\}\}_\gamma$ is the natural default for the troposphere.

8 Stability of MIS methods for this split

For a linear problem the frozen split with exact fast stages is MIS with $L = \partial F/\partial z$ and $N - L = \partial F/\partial y$. Linearized about a hydrostatic state with uniform wind $\bar u$, both operators are invariant under shifts of the columns, and one horizontal Fourier mode gives a block over the whole column (`MethodDesign/app_stability.jl`). On the gravity-wave grid ($\Delta x = 2.5\,\text{km}$, $n_z = 16$) the methods that are stable on the one-dimensional acoustic-advection test of Knoth and Wensch grow by 5–25 % per step already from $C_{\text{adv}} = \bar u\Delta t/\Delta x \approx 0.4$: the coupling of Section 5, item 4, is not present in that test. Methods designed on the column blocks (`MethodDesign/search_app.jl`):

method	frozen order	$\max(\rho-1)$, $C_{\text{adv}} \le 1.6$, $\bar u = 10/20/40$ m/s
MIS54 (Bauer & Knoth)	2	0.16 / 0.18 / 0.10
MISF53 (MIS54 projected onto frozen order 3)	3	0.14 / 0.26 / 0.26
MIS_FV42 (designed, from MIS4)	2	$7\cdot10^{-4}$ / $7\cdot10^{-3}$ / $7\cdot10^{-4}$
MIS_FV53 (designed, from MISF53)	3	$\le 6\cdot10^{-4}$

Nonlinear gravity wave (300 km channel, $U = 20\,\text{m/s}$, `:hevi`): at $T = 3000\,\text{s}$ all are accurate (MIS_FV53: wave error $1.2\cdot10^{-4}$ K at $\Delta t = 130\,\text{s}$, $1.8\cdot10^{-4}$ K at 200 s; MIS54: $1.6\cdot10^{-4}$, $1.6\cdot10^{-4}$); in a long run (36000 s at $\Delta t = 200\,\text{s}$, $C_{\text{adv}} = 1.6$) only MIS_FV53 survives. The frozen order (the bicoloured B-series conditions, `MethodDesign/BiColourMIS.jl`) is 2 for MIS54 and 3 for MISF53 and MIS_FV53.

9 Open points

- The split is implemented on flat grids only; terrain (Gal-Chen metric terms in the pressure gradient) and cut cells need the frozen coefficients of their metric and aperture terms.
- The well-balanced gravity mean of Section 7 is not implemented.

- Consistent advection: freezing the transported θ splits the Θ advection between the fast and the slow part; a split that keeps the advection of ρ and Θ together (e.g. also freezing the advective part of the mass flux) would remove the coupling that limits MIS stability.
- The DG counterpart: `docs/frozen_dg/frozen_dg.tex`.